MPC-MAP Assignment 4

Week 5: Kalman Filter

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Task 1 – Preparation

The robot stays still for 120 iterations, GNSS measurements are collected, the initial position is set to the sample mean and the GNSS covariance matrix is computed by `cov(gnss)`.

Task 2 – EKF implementation

The prediction step is implemented in `ekf_predict.m`. The state is $\mathbf{x} = [x, y, \theta]^T$ and the control is given by the wheel velocities. The differential-drive model is

$$\begin{aligned} x_k &= x_{k-1} + v\cos(\theta_{k-1})\Delta t \\ y_k &= y_{k-1} + v\sin(\theta_{k-1})\Delta t \\ \theta_k &= \theta_{k-1} + \omega\Delta t \end{aligned}$$

with the Jacobian

$$G_k = \begin{bmatrix} 1 & 0 & -v\sin(\theta)\Delta t \\ 0 & 1 & v\cos(\theta)\Delta t \\ 0 & 0 & 1 \end{bmatrix}.$$

The correction step is implemented in `kf_measure.m`. GNSS measures only position, therefore

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Task 3 – Filter tuning with known initial pose

The initial belief was set to $\mu_0 = [2, 2, \pi/2]^T$ and $\Sigma_0 = 0$. The process covariance was tuned as

$$R = \mathrm{diag}(0.01, 0.01, 0.01).$$

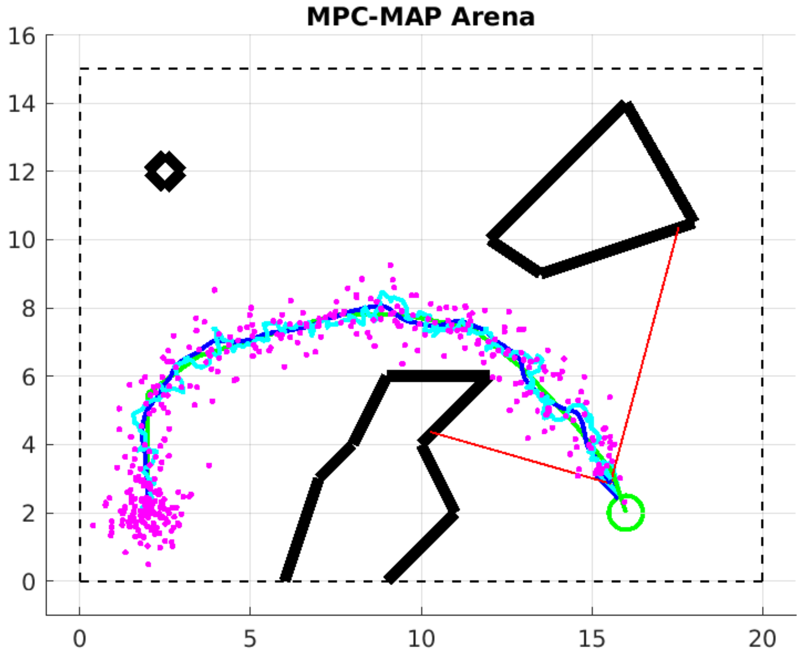

Figure 1: Known initial pose, EKF estimate and trajectory on `outdoor_1`.

Task 4 – Algorithm deployment

The initial position was obtained from the GNSS sample mean. The orientation was initialized with high variance because it is not measured directly:

$$\Sigma_0 = \begin{bmatrix} \Sigma_{GNSS} & 0 \\ 0 & 4 \end{bmatrix}.$$

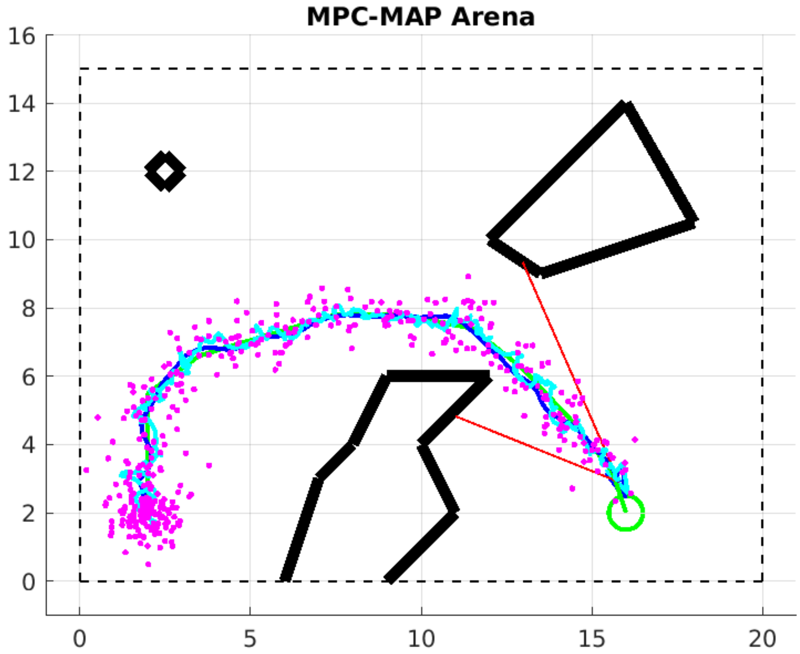

Figure 2: Unknown initial position from GNSS initialization and EKF ride to the goal.